

# Student Performance Prediction & Classification System

AI-Powered Academic Performance Report

## Student Information

|                    |          |                 |             |
|--------------------|----------|-----------------|-------------|
| Name               | Adnan    | Roll No         | 27600122237 |
| Gender             | Male     | Age             | 22          |
| Attendance         | 55.0%    | Internal Marks  | 65.0/100    |
| Assignment Score   | 45.0/100 | Study Hours/day | 8.0         |
| Previous Sem Marks | 65.0/100 |                 |             |

## Prediction Result

|                       |            |                  |                        |
|-----------------------|------------|------------------|------------------------|
| Predicted Grade       | C          | Result           | Pass                   |
| Category              | Average    | Confidence Score | 47.36%                 |
| Estimated Final Marks | 56.83/100  | Algorithm Used   | Random Forest          |
| Prediction Time       | 4499.56 ms | Generated On     | 12 Jul 2026, 16:09 UTC |

## Explainable AI — Why This Prediction?

| Factor                  | Contribution |           |
|-------------------------|--------------|-----------|
| Attendance              | 43.5%        | ■■■■■■■■■ |
| Study Hours             | 29.7%        | ■■■■■     |
| Assignment Score        | 13.0%        | ■■        |
| Internal Marks          | 9.0%         | ■         |
| Age                     | 2.6%         | ■         |
| Previous Semester Marks | 2.2%         | ■         |

## Personalized Recommendations

- **Attendance is below 75% (55%).** — Increase attendance to improve final performance. Aim for at least 85%.
- **Assignment score is low (45/100).** — Complete all pending assignments before the next exam cycle.

Generated automatically by the Student Performance Prediction & Classification System. This report is AI-generated for academic guidance purposes and should be reviewed by faculty.